Neap

Trial Examination 2025

HSC Year 12 Chemistry

General Instructions	• Reading time – 5 minutes • Working time – 3 hours • Write using black pen • Draw diagrams using pencil • Calculators approved by NESA may be used • A formulae sheet, data sheet and Periodic Table are provided at the back of this paper
Total Marks: 100	**Section I – 20 marks (pages 2–9)** • Attempt Questions 1–20 • Allow about 35 minutes for this section **Section II – 80 marks (pages 11–28)** • Attempt Questions 21–34 • Allow about 2 hours and 25 minutes for this section

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2025 HSC Year 12 Chemistry examination.

SECTION I

20 marks
Attempt Questions 1–20
Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

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1 Which of the following equations represents the reaction between solid calcium hydroxide and a hydrochloric acid solution?

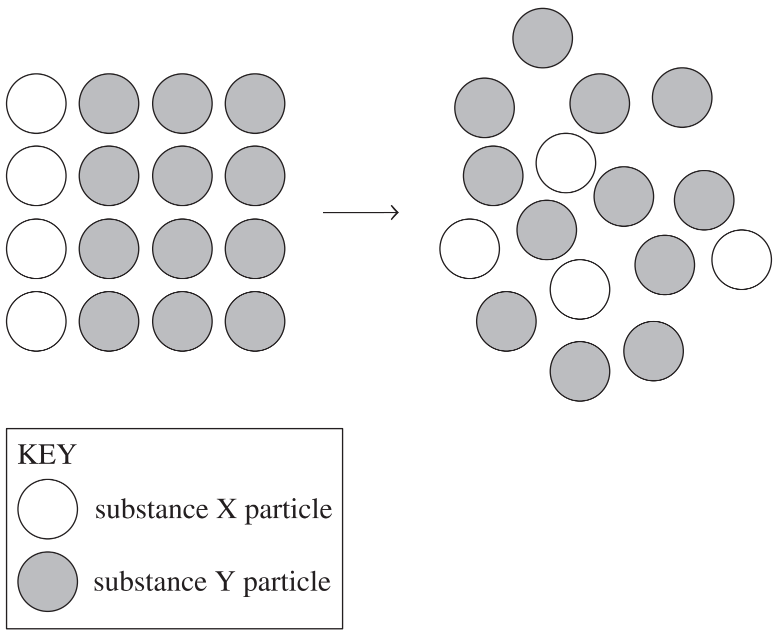
A. $Ca(OH)_2(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$

B. $Ca(OH)_2(aq) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$

C. $CaOH(s) + HCl(aq) \rightarrow CaCl(aq) + H_2O(l)$

D. $Ca(OH)_2(s) + HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l)$

2 The diagram represents the behaviour of particles of two substances, X and Y.



The diagram shows

A. substances X and Y bonding to form a new substance.

B. entropy increasing from left to right.

C. entropy decreasing from left to right.

D. the concentration of substance Y decreasing.

3 The equation represents an equilibrium reaction.

$$N_2O_4(g) + O_3(g) \rightleftharpoons N_2O_5(s) + O_2(g)$$

What is the equilibrium expression, K_{eq}, for the reaction?

A. $K_{eq} = \dfrac{N_2O_4 + O_3}{N_2O_5 + O_2}$

B. $K_{eq} = \dfrac{N_2O_5 + O_2}{N_2O_4 + O_3}$

C. $K_{eq} = \dfrac{[N_2O_4][O_3]}{[N_2O_5][O_2]}$

D. $K_{eq} = \dfrac{[O_2]}{[N_2O_4][O_3]}$

4 In an experiment, a strip of magnesium burns in air exothermically, according to the equation shown.

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

Which of the following statements is correct?

A. The enthalpy change, ΔH, of the reaction is positive.

B. The reaction is reversible.

C. The system is a non-equilibrium system.

D. The system is closed.

5 Titanium reacts with chlorine gas to form titanium tetrachloride, according to the equation shown.

$$Ti(s) + 2Cl_2(g) \rightleftharpoons TiCl_4(g)$$

Assuming that the system is at equilibrium, which of the following changes would shift the equilibrium position to the left?

A. adding a catalyst

B. adding $TiCl_4(g)$

C. removing $Ti(s)$

D. increasing the pressure

6 In an experiment, 0.20 mol of nitrosyl chloride gas was placed into a 1 L container. The container was sealed and the system was allowed to reach equilibrium as represented by the following equation.

$$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$$

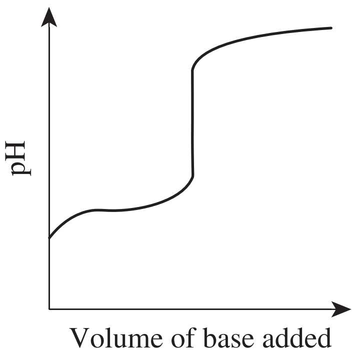
When the system was at equilibrium, there was 0.18 mol of NOCl present in the container.

What amounts of NO and Cl_2 were present at equilibrium?

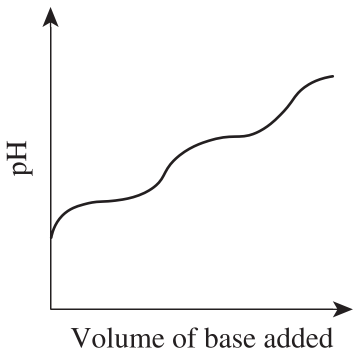
	NO(g) (mol)	*$Cl_2(g)$ (mol)*
A.	0.010	0.010
B.	0.010	0.020
C.	0.020	0.010
D.	0.020	0.020

7 Which of the following titration curves best represents the pH changes that occur when a strong base is added to a weak diprotic acid?

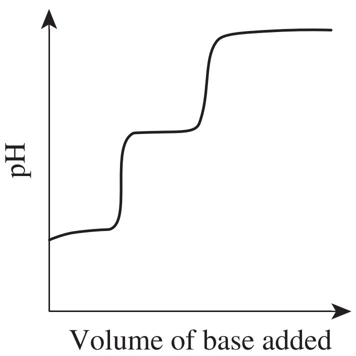
A.



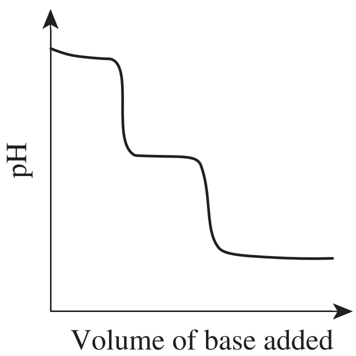
B.



C.



D.



8 Consider the acid/base reactions that are represented by the equations shown.

$$HCO_3^{-}(aq) + H_3O^{+}(aq) \rightarrow H_2CO_3(aq) + H_2O(l)$$

$$HCO_3^{-}(aq) + OH^{-}(aq) \rightarrow CO_3^{2-}(aq) + H_2O(l)$$

Based on these reactions, which species is amphiprotic?

A. $OH^{-}(aq)$

B. $HCO_3^{-}(aq)$

C. $CO_3^{2-}(aq)$

D. $H_2CO_3(aq)$

9 25.00 mL of a 0.10 mol L^{-1} solution of sodium hydroxide was diluted to 250.0 mL with distilled water.

What was the pH of the final solution?

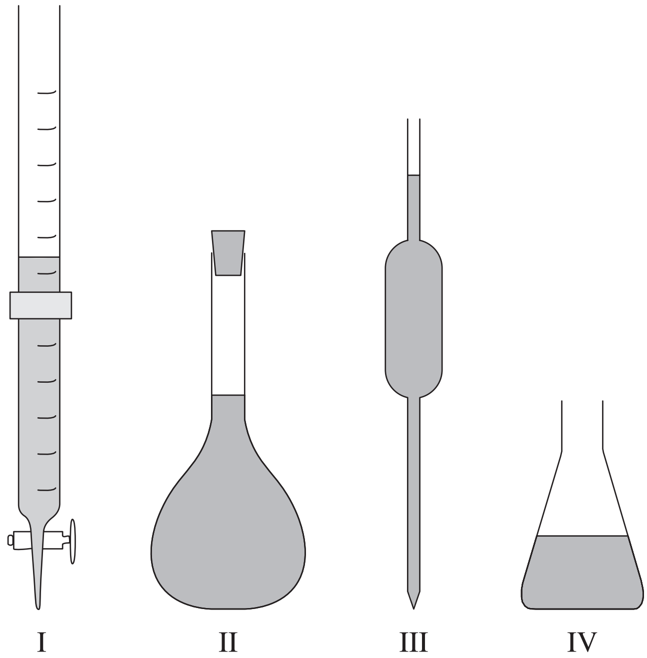
A. 1.0

B. 2.0

C. 11

D. 12

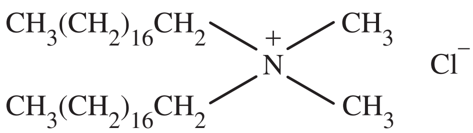
10 The diagrams illustrate four pieces of glassware (I, II, III and IV) that are used in titration.



Which row of the table identifies how each piece of glassware is used in a titration?

	I	*II*	*III*	*IV*
A.	used to deliver titrant	contains analyte and indicator	used to measure analyte	used to prepare standard solutions
B.	used to measure analyte	used to prepare standard solutions	used to deliver titrant	contains analyte and indicator
C.	used to deliver titrant	used to prepare standard solutions	used to measure analyte	contains analyte and indicator
D.	used to measure analyte	contains analyte and indicator	used to deliver titrant	used to prepare standard solutions

11 Consider the diagram.



What does the diagram represent?

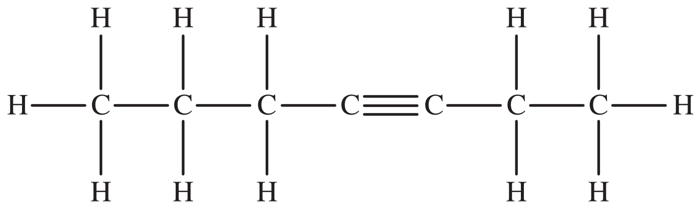
A. a soap molecule

B. an anionic detergent molecule

C. a cationic detergent molecule

D. a neutral surfactant molecule

12 The structural formula of a compound is shown.



What is the IUPAC name of this compound?

A. hept-3-yne

B. pent-3-yne

C. hept-3-ene

D. 1-ethyl-2-propylethyne

13 How many position isomers does $C_3H_6Br_2$ have?

A. 1

B. 2

C. 3

D. 4

14 Which of the following lists the compounds in order of decreasing boiling point?

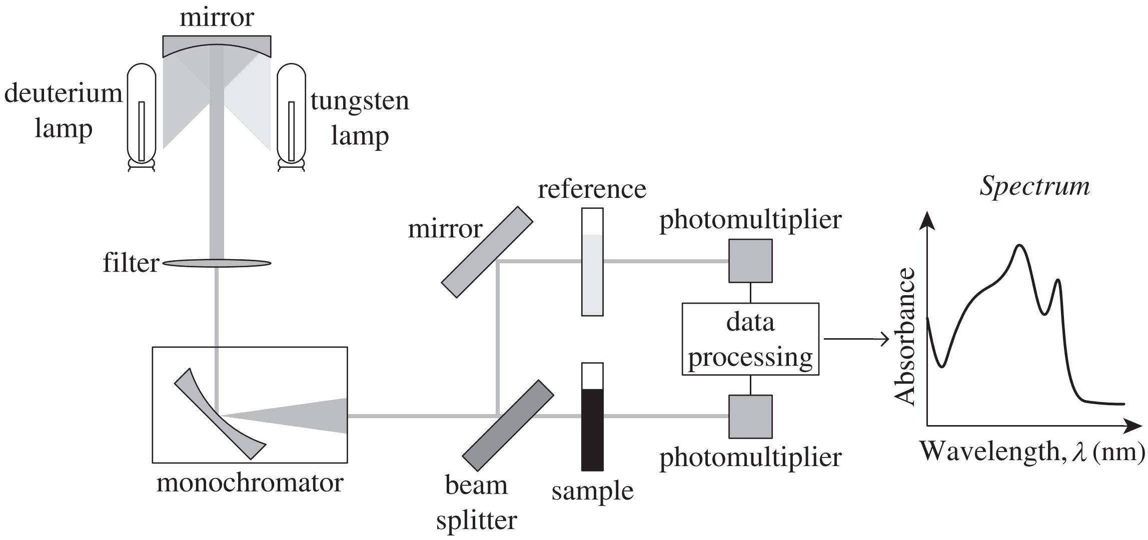
A. propane > ethanol > ethanoic acid > ethanamine

B. ethanoic acid > ethanol > ethanamine > propane

C. ethanoic acid > ethanol > propane > ethanamine

D. propane > ethanamine > ethanol > ethanoic acid

15 The diagram illustrates a particular analytical technique.



The table provides data relating to the light sources used.

Light source	*Wavelength (nm)*
deuterium lamp	$\lambda < 400$
tungsten lamp	$350 < \lambda < 1000$

The analytical technique illustrated is

A. ultraviolet-visible spectrophotometry.

B. atomic absorption spectroscopy.

C. mass spectroscopy.

D. infrared spectroscopy.

16 The ^{1}H NMR spectrum of bromoethane has two sets of peaks. Both peaks are split. Which of the following describes the splitting pattern in the spectrum?

A. a singlet and a doublet

B. a doublet and a doublet

C. a doublet and a triplet

D. a triplet and a quartet

17 Which of the following compounds is a functional group isomer of pentanal?

A. $CH_3CH_2CH_2COCH_3$

B. $CH_3CH_2CH_2CH_2CHO$

C. $CH_3CH_2CH_2COOCH_3$

D. $(CH_3)_2CHCH_2CHO$

18 Which of the following molecules has a linear shape?

A. ethene

B. ethyne

C. ethane

D. methanoic acid

19 Glycine (H_2NCH_2COOH) can be used to make a biodegradable condensation polymer.
The molecular mass of glycine is 75.07 g mol^{-1}.
What is the molar mass of the polymer formed from 1000 glycine monomer units?

A. 57.1 g mol^{-1}

B. 3.9×10^4 g mol^{-1}

C. 5.7×10^4 g mol^{-1}

D. 7.5×10^4 g mol^{-1}

20 Which of the following molecules will NOT react with acidified potassium permanganate?

A. 2-methylbutan-2-ol

B. pentanal

C. 2,2-dimethylpropan-1-ol

D. pentan-3-ol

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HSC Year 12 Chemistry

Section II Answer Booklet

80 marks
Attempt Questions 21–34
Allow about 2 hours and 25 minutes for this section

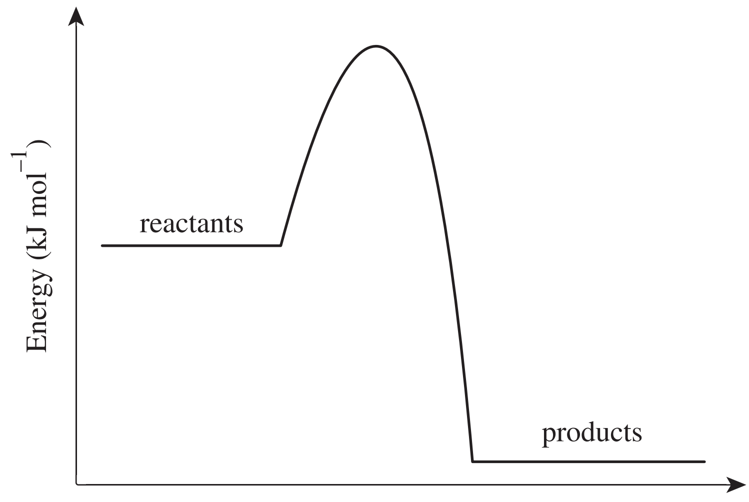
Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in questions involving calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

Question 21 (7 marks)

The incomplete energy profile diagram for an equilibrium reaction is shown.



(a) On the diagram, draw and label the activation energies, E_a, for the forward and reverse reactions, and the enthalpy change, ΔH. **3**

(b) With reference to the energy profile diagram, outline the significance of activation energy and enthalpy change in equilibrium reactions, and describe what would happen to this system if the temperature were increased. **4**

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Question 22 (4 marks)

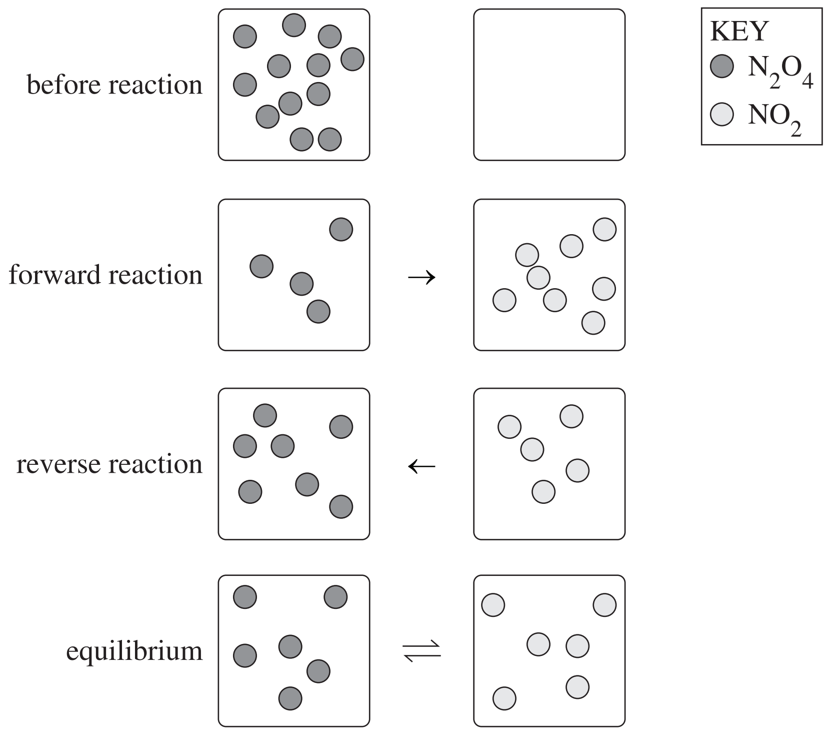
Using an example, describe how First Nations peoples remove toxins from food using solubility equilibria. **4**

Question 23 (9 marks)

A student's model of dynamic equilibrium is shown.

A model of dynamic equilibrium using the decomposition of dinitrogen tetroxide

The reaction occurs in a sealed container and according to the equation $N_2O_4(g) \rightleftharpoons 2NO_2(g)$.



Question 23 continues on page 15

Question 23 (continued)

(a) Evaluate the effectiveness of the student's model in demonstrating the features of dynamic equilibrium. **7**

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(b) For the system shown, how could the product yield be increased without adding or removing reagents? Justify your answer. **2**

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End of Question 23

Question 24 (8 marks)

(a) Using an example, outline how a natural indicator can be prepared and tested. **4**

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(b) Explain how the reversible reactions of indictors enable the identification of acids and bases. Include ONE balanced chemical equation in your answer. **4**

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Question 25 (8 marks)

A research assistant is tasked with making an acidic buffer solution using two solutions that each have a concentration of 0.1 M. The solutions that are available for the research assistant to use are shown in the table.

Solution name	*Formula*
methanoic acid	$HCOOH$
sodium methanoate	$HCOONa$
nitric acid	HNO_3
sodium nitrate	$NaNO_3$
sulfuric acid	H_2SO_4
potassium hydroxide	KOH

(a) Outline the function of a buffer solution. **1**

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(b) Identify the TWO solutions that the research assistant should use and explain how the acidic buffer solution will work. Include balanced chemical equations in your answer. **7**

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Question 26 (4 marks)

Using examples, explain the difference between the terms strong, weak, concentrated and dilute when referring to acids. **4**

Question 27 (4 marks)

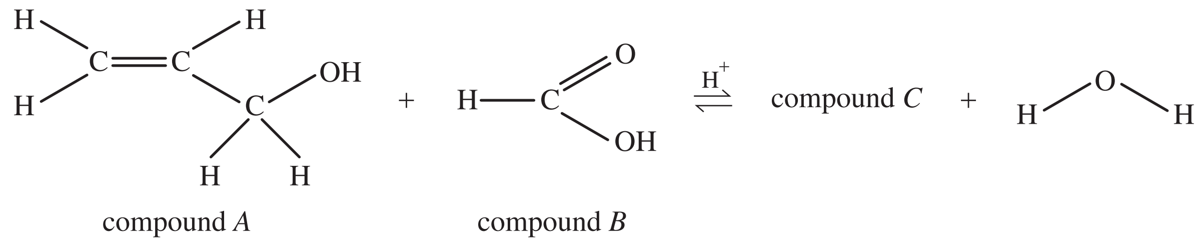
Glucose undergoes fermentation to produce ethanol.

(a) Write the balanced chemical equation for this reaction. **1**

(b) Outline an experiment that could be conducted to monitor mass changes during this reaction. **3**

Question 28 (6 marks)

Organic compound *A* reacts with compound *B* to produce compound *C*, as illustrated by the chemical equation. The reaction is exothermic and uses a sulfuric acid (H_2SO_4) catalyst.



(a) Identify TWO safe work practices that should be used when conducting this reaction in a school science laboratory. **2**

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(b) Draw the structural formula of compound *C*. **1**

(c) A polymer can be produced from compound *C*. **3**

Identify the type of polymer that can be produced and outline ONE possible method of its production. Include a structural formula of three monomer units of the polymer in your answer.

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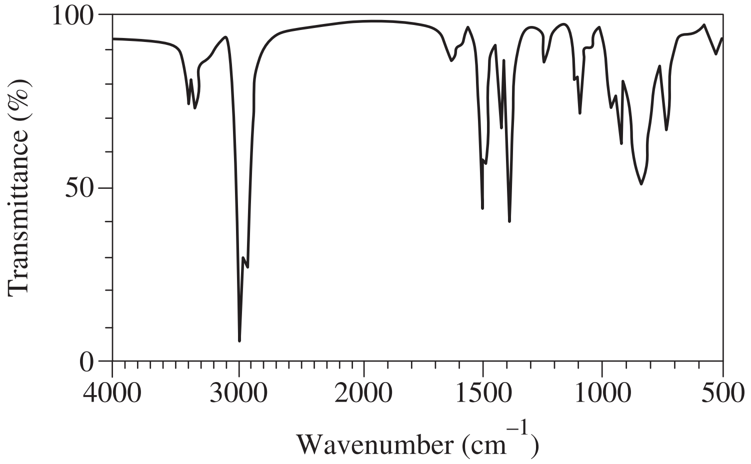
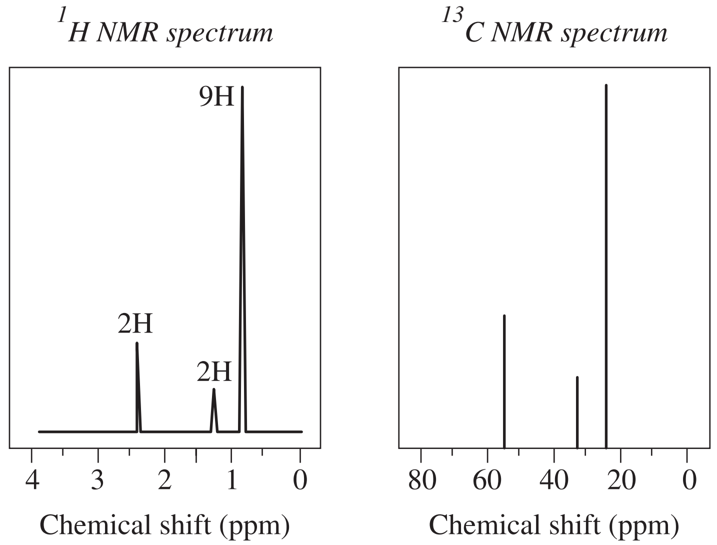
Question 29 (7 marks)

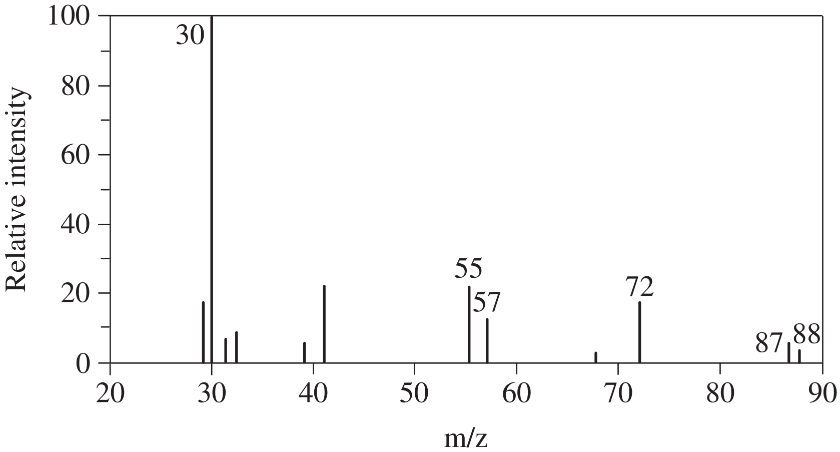
Combustion analysis of an unknown organic compound indicated that the compound contained carbon, hydrogen and nitrogen only. The spectra of the compound are shown. **7**

Infrared spectrum





Mass spectrum



Question 29 continues on page 21

Question 29 (continued)

Identify the unknown compound and draw its structural formula. Justify your answer with reference to the spectra provided.

Name ....

Structure

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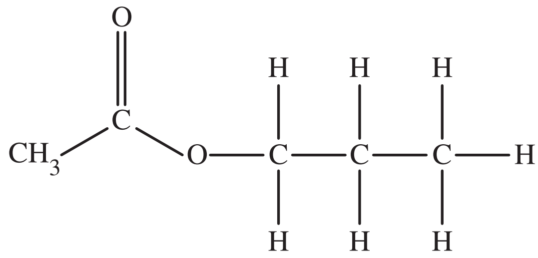
End of Question 29

Question 30 (6 marks)

Artificial pear essence is a mixture that comprises 50.0% water, 37.5% ethanol and 12.5% 1-propyl ethanoate.

(a) The structural formula of a molecule of 1-propyl ethanoate is shown. **3**

On the structural formula provided, draw water molecules forming hydrogen bonds with the 1-propyl ethanoate molecule.



(b) With reference to polarity and bonding, explain why ethanol is a suitable co-solvent for artificial pear essence. **3**

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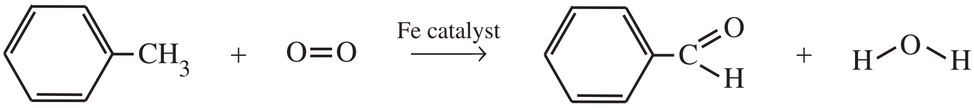
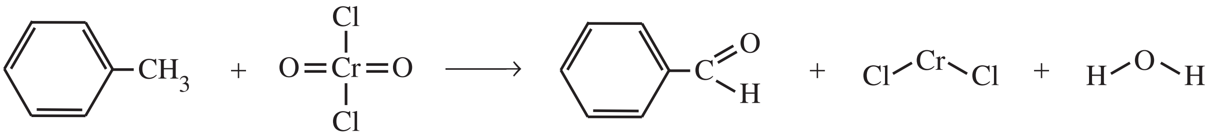
Question 31 (4 marks)

Two industrial processes that use toluene in the synthesis of benzaldehyde are represented by the equations shown. **4**

Process 1



Process 2



The table shows data relating to the processes.

	Temperature (°C)	*Pressure (kPa)*	*Percentage of toluene converted to benzaldehyde (%)*
Process 1	100	100	20
Process 2	60	100	93

Which process is preferable? Justify your answer with reference to the information provided.

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Question 32 (5 marks)

A group of campers used a butane gas stove to heat a kettle containing 2150 mL of water. The initial temperature of the water was 12.0°C. Only 80.0% of the heat released by the combustion of butane is absorbed by the water in the kettle.

(a) The standard enthalpy of combustion, ΔH_c, of butane is –2878 kJ mol^{-1}. **4**

Calculate the final temperature of the water if 18.0 g of butane is combusted.

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(b) Butane is a fossil fuel. **1**

Describe ONE environmental impact of obtaining and using fossil fuels.

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Question 33 (3 marks)

Explain the cleaning action of soap. Include a diagram in your answer. **3**

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Question 34 (5 marks)

A 1.00 mol L^{-1} solution contains at least one of the cations Ba^{2+}, Ca^{2+} and Ag^{+}. **5**

Outline a series of tests that could be performed in a school laboratory to determine the cation, or cations, present in the solution. Include expected observations and balanced chemical equations in your answer.

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FORMULAE SHEET

$n = \dfrac{m}{MM}$	$c = \dfrac{n}{V}$	$PV = nRT$
$q = mc\Delta T$	$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$	$\text{pH} = -\log_{10}\left[\text{H}^+\right]$
$pK_a = -\log_{10}\left[K_a\right]$	$A = \varepsilon lc = \log_{10}\dfrac{I_o}{I}$	

Avogadro constant, N_A		6.022×10^{23} mol^{-1}
Volume of 1 mole ideal gas:	at 100 kPa and	
	at 0°C (273.15 K)	22.71 L
	at 25°C (298.15 K)	24.79 L
Gas constant		8.314 J mol^{-1} K^{-1}
Ionisation constant for water at 25°C (298.15 K), K_w		1.0×10^{-14}
Specific heat capacity of water		4.18×10^{3} J kg^{-1} K^{-1}

DATA SHEET

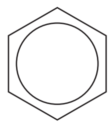
Solubility constants at 25°C

Compound	K_{sp}	*Compound*	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Infrared absorption data

Bond	Wavenumber/cm^{-1}
N—H (amines)	3300–3500
O—H (alcohols)	3230–3550 (broad)
C—H	2850–3300
O—H (acids)	2500–3000 (very broad)
C≡N	2220–2260
C═O	1680–1750
C═C	1620–1680
C—O	1000–1300
C—C	750–1100

^{13}C NMR chemical shift data

Type of carbon	δ/ppm
$-\overset{\vert}{\underset{\vert}{C}}-\overset{\vert}{\underset{\vert}{C}}-$	5–40
$R-\overset{\vert}{\underset{\vert}{C}}-$ Cl or Br	10–70
$R-\underset{\Vert}{\overset{}{C}}-\overset{\vert}{\underset{\vert}{C}}-$ (C═O)	20–50
$R-\overset{\vert}{\underset{\vert}{C}}-N\langle$	25–60
$-\overset{\vert}{\underset{\vert}{C}}-O-$ alcohols, ethers or esters	50–90
$\rangle C=C\langle$	90–150
$R-C\equiv N$	110–125
	110–160
$R-\underset{\Vert}{C}-$ (C═O) esters or acids	160–185
$R-\underset{\Vert}{C}-$ (C═O) aldehydes or ketones	190–220

UV absorption

(This is not a definitive list and is approximate.)

Chromophore	λ_{max} (nm)
C—H	122
C—C	135
C═C	162

Chromophore	λ_{max} (nm)
C≡C	173 178 196 222
C—Cl	173
C—Br	208

 HSC_Chem_Y12_QB_2025

Some standard potentials

$K^+ + e^-$	$\rightleftharpoons$	$K(s)$	–2.94 V
$Ba^{2+} + 2e^-$	$\rightleftharpoons$	$Ba(s)$	–2.91 V
$Ca^{2+} + 2e^-$	$\rightleftharpoons$	$Ca(s)$	–2.87 V
$Na^+ + e^-$	$\rightleftharpoons$	$Na(s)$	–2.71 V
$Mg^{2+} + 2e^-$	$\rightleftharpoons$	$Mg(s)$	–2.36 V
$Al^{3+} + 3e^-$	$\rightleftharpoons$	$Al(s)$	–1.68 V
$Mn^{2+} + 2e^-$	$\rightleftharpoons$	$Mn(s)$	–1.18 V
$H_2O + e^-$	$\rightleftharpoons$	$\frac{1}{2} H_2(g) + OH^-$	–0.83 V
$Zn^{2+} + 2e^-$	$\rightleftharpoons$	$Zn(s)$	–0.76 V
$Fe^{2+} + 2e^-$	$\rightleftharpoons$	$Fe(s)$	–0.44 V
$Ni^{2+} + 2e^-$	$\rightleftharpoons$	$Ni(s)$	–0.24 V
$Sn^{2+} + 2e^-$	$\rightleftharpoons$	$Sn(s)$	–0.14 V
$Pb^{2+} + 2e^-$	$\rightleftharpoons$	$Pb(s)$	–0.13 V
$H^+ + e^-$	$\rightleftharpoons$	$\frac{1}{2} H_2(g)$	0.00 V
$SO_4^{2-} + 4H^+ + 2e^-$	$\rightleftharpoons$	$SO_2(aq) + 2H_2O$	0.16 V
$Cu^{2+} + 2e^-$	$\rightleftharpoons$	$Cu(s)$	0.34 V
$\frac{1}{2} O_2(g) + H_2O + 2e^-$	$\rightleftharpoons$	$2OH^-$	0.40 V
$Cu^+ + e^-$	$\rightleftharpoons$	$Cu(s)$	0.52 V
$\frac{1}{2} I_2(s) + e^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2} I_2(aq) + e^-$	$\rightleftharpoons$	I^-	0.62 V
$Fe^{3+} + e$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$Ag^+ + e^-$	$\rightleftharpoons$	$Ag(s)$	0.80 V
$\frac{1}{2} Br_2(l) + e^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2} Br_2(aq) + e^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2} O_2(g) + 2H^+ + 2e^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2} Cl_2(g) + e^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2} Cr_2O_7^{2-} + 7H^+ + 3e^-$	$\rightleftharpoons$	$Cr^{3+} + \frac{7}{2} H_2O$	1.36 V
$\frac{1}{2} Cl_2(aq) + e^-$	$\rightleftharpoons$	Cl^-	1.40 V
$MnO_4^- + 8H^+ + 5e^-$	$\rightleftharpoons$	$Mn^{2+} + 4H_2O$	1.51 V
$\frac{1}{2} F_2(g) + e^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data (5th Edition)* is the principal source of data for the standard potentials. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

KEY

atomic number	79
symbol	Au
standard atomic weight	197.0
name	gold

1 H 1.008 hydrogen																	2 He 4.003 helium
3 Li 6.941 lithium	4 Be 9.012 beryllium											5 B 10.81 boron	6 C 12.01 carbon	7 N 14.01 nitrogen	8 O 16.00 oxygen	9 F 19.00 fluorine	10 Ne 20.18 neon
11 Na 22.99 sodium	12 Mg 24.31 magnesium											13 Al 26.98 aluminium	14 Si 28.09 silicon	15 P 30.97 phosphorus	16 S 32.07 sulfur	17 Cl 35.45 chlorine	18 Ar 39.95 argon
19 K 39.10 potassium	20 Ca 40.08 calcium	21 Sc 44.96 scandium	22 Ti 47.87 titanium	23 V 50.94 vanadium	24 Cr 52.00 chromium	25 Mn 54.94 manganese	26 Fe 55.85 iron	27 Co 58.93 cobalt	28 Ni 58.69 nickel	29 Cu 63.55 copper	30 Zn 65.38 zinc	31 Ga 69.72 gallium	32 Ge 72.64 germanium	33 As 74.92 arsenic	34 Se 78.96 selenium	35 Br 79.90 bromine	36 Kr 83.80 krypton
37 Rb 85.47 rubidium	38 Sr 87.61 strontium	39 Y 88.91 yttrium	40 Zr 91.22 zirconium	41 Nb 92.91 niobium	42 Mo 95.96 molybdenum	43 Tc technetium	44 Ru 101.1 ruthenium	45 Rh 102.9 rhodium	46 Pd 106.4 palladium	47 Ag 107.9 silver	48 Cd 112.4 cadmium	49 In 114.8 indium	50 Sn 118.7 tin	51 Sb 121.8 antimony	52 Te 127.6 tellurium	53 I 126.9 iodine	54 Xe 131.3 xenon
55 Cs 132.9 caesium	56 Ba 137.3 barium	57–71 lanthanoids	72 Hf 178.5 hafnium	73 Ta 180.9 tantalum	74 W 183.9 tungsten	75 Re 186.2 rhenium	76 Os 190.2 osmium	77 Ir 192.2 iridium	78 Pt 195.1 platinum	79 Au 197.0 gold	80 Hg 200.6 mercury	81 Tl 204.4 thallium	82 Pb 207.2 lead	83 Bi 209.0 bismuth	84 Po polonium	85 At astatine	86 Rn radon
87 Fr francium	88 Ra radium	89–103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganesson

Lanthanoids

57 La 138.9 lanthanum	58 Ce 140.1 cerium	59 Pr 140.9 praseodymium	60 Nd 144.2 neodymium	61 Pm promethium	62 Sm 150.4 samarium	63 Eu 152.0 europium	64 Gd 157.3 gadolinium	65 Tb 158.9 terbium	66 Dy 162.5 dysprosium	67 Ho 164.9 holmium	68 Er 167.3 erbium	69 Tm 168.9 thulium	70 Yb 173.1 ytterbium	71 Lu 175.0 lutetium

Actinoids

89 Ac actinium	90 Th 232.0 thorium	91 Pa 231.0 protactinium	92 U 238.0 uranium	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium

Standard atomic weights are abridged to four significant figures.
Elements with no reported values in the table have no stable nuclides.
Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

Neap

Trial Examination 2025

HSC Year 12 Chemistry

Solutions and Marking Guidelines

SECTION I

Question 1 A

A is correct. Calcium hydroxide ($Ca(OH)_2(s)$) is a base and will undergo an acid/base reaction with a solution of hydrochloric acid ($HCl(aq)$) to produce the salt calcium chloride ($CaCl_2(aq)$) and water. The stoichiometric ratio of $Ca(OH)_2(s)$ and $HCl(aq)$ in the reaction is 1 : 2.

B is incorrect. This option shows $Ca(OH)_2$ as a solution, not a solid.

C is incorrect. This option uses incorrect formulae for calcium hydroxide and calcium chloride.

D is incorrect. This option uses incorrect ratios for HCl and water.

Mod 6 Acid/Base Reactions (CH12–5)	Bands 2–3

Question 2 B

B is correct and **C** is incorrect. Entropy refers to the level of randomness or disorder in a system. From left to right, the diagram shows the particles becoming disordered. Hence, it shows entropy increasing from left to right.

A is incorrect. Bonding between the particles is not indicated in the diagram.

D is incorrect. The concentrations of both substances remain the same; there are four particles of substance X and 12 particles of substance Y.

Mod 5 Equilibrium and Acid Reactions (CH12–6)	Bands 2–3

Question 3 D

For a reaction in the form $aA + bB \rightleftharpoons cC + dD$, the equilibrium expression is:

$$K_{eq} = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Since N_2O_5 is a solid, its activity is 1. Therefore, the equilibrium expression for the reaction represented by the equation $N_2O_4(g) + O_3(g) \rightleftharpoons N_2O_5(s) + O_2(g)$ is:

$$K_{eq} = \frac{[O_2]}{[N_2O_4][O_3]}$$

Mod 5 Equilibrium and Acid Reactions (CH12–6, 12–12)	Bands 3–4

Question 4 C

C is correct and **B** is incorrect. An equilibrium reaction is a reversible reaction and is indicated by the equilibrium symbol ($\rightleftharpoons$). The equation shown uses a right arrow ($\rightarrow$), which indicates a non-equilibrium, and hence irreversible, reaction.

A is incorrect. Exothermic reactions have negative enthalpy change, ΔH, values. Endothermic reactions have positive ΔH values.

D is incorrect. As the magnesium burns in air, energy and matter can leave the system. Therefore, the system is open.

Mod 5 Equilibrium and Acid Reactions (CH12–6, 12–12)	Bands 3–4

Question 5 **B**

B is correct. When a system is at equilibrium, the addition of a product will shift the equilibrium position to the left to form more reactants.

A is incorrect. Catalysts increase reaction rate, but do not affect the position of equilibrium.

C is incorrect. Removing or adding solids to an equilibrium mixture does not change the equilibrium position.

D is incorrect. Increasing the pressure of a system with gases present at equilibrium causes the equilibrium position to shift in the direction that produces fewer gaseous molecules, which decreases pressure. In this case, the forward reaction produces one gaseous molecule and the reverse reaction produces two gaseous molecules. Therefore, increasing the pressure would shift the equilibrium position to the right.

Mod 5 Equilibrium and Acid Reactions (CH12–6, 12–12)	Bands 3–4

Question 6 **C**

	$NOCl(g)$	$2NO(g)$	$Cl_2(g)$
$n_{initial}$	0.20	0.0	0.0
n_{change}	–0.020	+0.020	+0.010
$n_{equilibrium}$	0.18	0.020	0.010

Mod 5 Equilibrium and Acid Reactions (CH12–6)	Bands 5–6

Question 7 **B**

B is correct. A diprotic acid can donate two protons (H^+) and dissociates in two steps. When a strong base is added to a weak diprotic acid, the pH is initially less than 7 and greater than 3, then increases in two gradual steps.

A is incorrect. This option represents the changes that occur when a strong base is added to a weak monoprotic acid.

C is incorrect. This option represents the changes that occur when a strong base is added to a strong diprotic acid.

D is incorrect. This option represents the changes that occur when a strong diprotic acid is added to a strong base.

Mod 6 Acid/Base Reactions (CH12–6, 12–13)	Bands 5–6

Question 8 **B**

B is correct. An amphiprotic species can donate a proton (H^+) or accept an additional proton. In the first reaction, $HCO_3^-(aq)$ accepts a proton. In the second reaction, it donates a proton.

A is incorrect. In the second reaction, $OH^-(aq)$ accepts a proton.

C is incorrect. In the second reaction, $CO_3^{2-}(aq)$ can accept a proton only.

D is incorrect. In the first reaction, $H_2CO_3(aq)$ can donate a proton only.

Mod 6 Acid/Base Reactions (CH12–6, 12–13)	Bands 4–5

Question 9 D

The original solution was diluted by a factor of $\frac{250.0}{25.00} = 10,$ so the concentration of the final solution was 0.010 mol L^{-1}. Therefore:

$$\begin{aligned} pOH &= -\log\left[OH^-\right] \\ &= -\log[0.010] \\ &= 2.0 \\ pH + pOH &= 14 \\ pH &= 14 - 2.0 \\ &= 12 \end{aligned}$$

Mod 6 Acid/Base Reactions (CH12–6, 12–13)	Bands 4–5

Question 10 C

In titration, the titrant is a standard solution (that is, a solution of known concentration) and is added to the analyte, which is the solution whose concentration is to be determined. A bulb pipette (III) is used to measure a fixed volume of analyte, which is then added to a conical flask (IV). An indicator is also added to the flask. The titrant is prepared in a volumetric flask (II) and then delivered into the conical flask via a burette (I).

Mod 6 Acid/Base Reactions (CH12 -6, 12–13)	Bands 5–6

Question 11 C

C is correct and **D** is incorrect. The diagram represents an ammonium salt, which is a detergent (or surfactant). Ammonium salts are cationic (positively charged) molecules.

A is incorrect. Soap molecules have the formula $RCOO^-Na^+$.

B is incorrect. Anionic detergents (or surfactants) have a negatively charged head group, often a sulfonate group with the formula $R–SO_3^-Na^+$.

Mod 7 Organic Chemistry (CH12–6, 12–7, 12–14)	Bands 4–5

Question 12 A

A is correct. There are seven carbon atoms in the longest carbon chain; therefore, the base is ‘hept-’. The functional group is an alkyne $(C \equiv C)$; therefore, the suffix is ‘-yne’. The first carbon of the alkyne functional group is carbon 3 counting from the end of the molecule that is closest to the group; therefore, the location number of the functional group is ‘3’. Hence, the IUPAC name is hept-3-yne.

B is incorrect. The base ‘pent-’ indicates five carbon atoms.

C is incorrect. The suffix ‘-ene’ indicates an alkene functional group (C=C).

D is incorrect. This option does not identify the longest carbon chain.

Mod 7 Organic Chemistry (CH12–4, 12–5, 12–7)	Bands 4–5

Question 13 D

The position isomers of $C_3H_6Br_2$ are 1,1-dibromopropane; 1,2-dibromopropane; 1,3-dibromopropane; and 2,2-dibromopropane.

Mod 7 Organic Chemistry (CH12–5, 12–6, 12–7, 12–14)	Bands 5–6

Question 14 B

The compounds are all small molecules of similar size and so the strengths of their dispersion forces are similar. Propane is non-polar and so has only dispersion forces between its molecules, meaning that it has the lowest boiling point.

Ethanamine has a nitrogen atom and so has dipole–dipole forces and hydrogen bonding between its molecules. Therefore, it has a higher boiling point than propane.

Ethanol has an oxygen atom and so has dipole–dipole forces and hydrogen bonding between its molecules. Oxygen is more electronegative than nitrogen, meaning that ethanol has a higher boiling point than ethanamine.

Ethanoic acid has two very electronegative oxygen atoms and so is the most polar molecule, meaning that it has the strongest dipole–dipole forces and strongest hydrogen bonding. Therefore, it has the highest boiling point.

Mod 7 Organic Chemistry (CH12–4, 12–5, 12–14)	Bands 5–6

Question 15 A

A is correct. Ultraviolet-visible spectrophotometry requires two light sources. The deuterium lamp provides ultraviolet radiation, and the tungsten lamp provides visible radiation.

B is incorrect. Atomic absorption spectroscopy uses hollow cathode tubes as light sources.

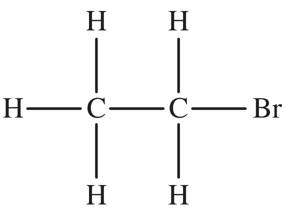
C is incorrect. In mass spectroscopy, ions pass through a magnetic field. A light source is not required.

D is incorrect. Infrared spectroscopy uses a low energy thermal lamp that produces light with wavelengths between 1000 nm and 1 mm.

Mod 8 Applying Chemical Ideas (CH12–3, 12–7, 12–15)	Bands 4–5

Question 16 D

The structural formula of bromoethane is shown.



The *n* + 1 rule can be used to determine the splitting patterns in first order NMR spectra. The CH_2 is split by three adjacent protons into a quartet. The CH_3 is split by two adjacent protons into a triplet.

Mod 8 Applying Chemical Ideas (CH12–6, 12–14, 12–15)	Bands 5–6

Question 17 **A**

A is correct. This option is pentan-2-one, which is a functional group isomer of pentanal.

B is incorrect. This option is pentanal.

C is incorrect. This option is methyl butanoate, which is not an isomer of pentanal.

D is incorrect. This option is 2-methylbutanal, which is a chain isomer of pentanal.

Mod 7 Organic Chemistry (CH12–5, 12–7, 12–14)	
Mod 8 Applying Chemical Ideas	Bands 4–5

Question 18 **B**

B is correct. The structural formula of ethyne is shown.

$H-C\equiv C-H$

A is incorrect. An ethene molecule has a trigonal planar shape.

C is incorrect. An ethane molecule has a tetrahedral shape.

D is incorrect. A methanoic acid molecule has a trigonal planar shape.

Mod 7 Organic Chemistry (CH12–5, 12–7, 12–14)	Bands 3–4

Question 19 **C**

The equation for the polymerisation of glycine shows that one molecule of water is lost for every molecule of glycine: $n \times H_2NCH_2COOH \rightarrow -\left(HNCH_2CO\right)_n + n\left(H_2O\right)$.

Therefore:

$$1000 \times 75.07 \rightarrow \text{polymer} + 1000 \times 18$$

$$\text{polymer} = 1000 \times 75.07 - 1000 \times 18.00$$

$$= 5.7 \times 10^4 \text{ g mol}^{-1}$$

Mod 7 Organic Chemistry (CH12–6, 12–14)	Bands 5–6

Question 20 **A**

A is correct. 2-methylbutan-2-ol is a tertiary alcohol. Tertiary alcohols resist oxidation and so will not react with acidified potassium permanganate.

B is incorrect. Pentanal will be oxidised by acidified potassium permanganate to produce pentanoic acid.

C is incorrect. 2,2-dimethylpropan-1-ol is a primary alcohol and will be oxidised by acidified potassium permanganate to produce 2,2-dimethylpropanoic acid.

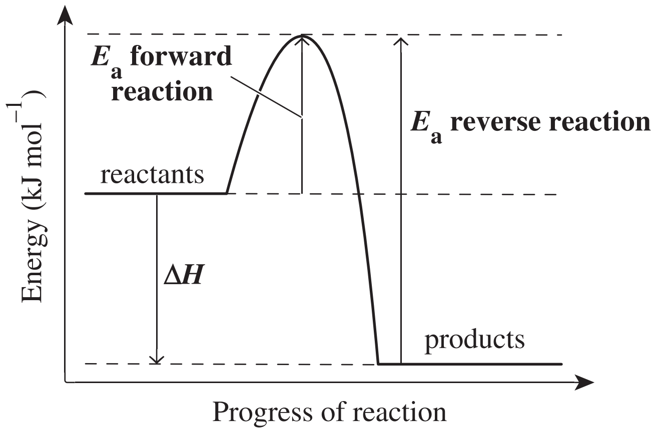
D is incorrect. Pentan-3-ol is a secondary alcohol and will be oxidised by acidified potassium permanganate to produce pentan-3-one.

Mod 7 Organic Chemistry (CH12–5, 12–6, 12–14)	Bands 5–6

SECTION II

Question 21

(a)



Mod 5 Equilibrium and Acid Reactions (CH12–6, 12–12)	Bands 3–4
• Draws and labels – the E_a for the forward reaction – the E_a for the reverse reaction – the ΔH	3
• Any TWO of the above points	2
• Any ONE of the above points	1

(b) *For example:*

Activation energy, E_a, is the minimum energy required to cause a chemical reaction. In this case, the forward reaction has a lower activation energy than the reverse reaction, and so it will reach equilibrium faster than the reverse reaction.

Enthalpy change, ΔH, is the amount of energy released or absorbed during a reaction. A negative enthalpy change indicates that a reaction releases energy and so is exothermic. A positive enthalpy change indicates that a reaction absorbs energy and so is endothermic. In this case, the forward reaction is exothermic and the reverse reaction is endothermic. If the temperature of the reaction were decreased, the equilibrium position would shift to the right to favour the forward reaction. If the temperature of the reaction were increased, the equilibrium position would shift to the left to favour the reverse reaction.

Mod 5 Equilibrium and Acid Reactions (CH12–12)	Bands 5–6
• Outlines E_a and ΔH. AND • Outlines the significance of E_a in terms of equilibrium reactions. AND • Outlines the significance of ΔH in terms of equilibrium reactions. AND • Describes what would happen if the temperature were increased	4
• Any THREE of the above points	3
• Any TWO of the above points	2
• Provides some relevant information	1

Question 22

For example:

Aboriginal peoples have used their knowledge of solubility equilibria to remove toxins from the seeds of *Macrozamia* fruit. One Indigenous knowledge practice involves placing the seeds in shallow lakes for several days to leach the water-soluble toxin. In a closed system, the reaction would reach equilibrium and some of the toxin would remain undissolved in the seed. However, the reaction is carried out in an open system where the water is flowing. This means that the dissolved toxin is continuously removed and equilibrium is not reached, resulting in the toxin being removed from the seeds completely.

Mod 5 Equilibrium and Acid Reactions (CH12–7, 12–12)	Bands 5–6
• Uses an appropriate example. AND • Describes the Indigenous knowledge practice. AND • Describes the principles used in the practice	4
• Uses an appropriate example. AND • Describes the Indigenous knowledge practice. AND • States the principles used in the practice	3
• Uses an appropriate example. AND • Describes the Indigenous knowledge practice.	2
• Provides some relevant information	1

Question 23

(a) *For example:*

The student's model uses an appropriate reaction and the correct chemical equation. It clearly shows the reactant, product, forward reaction, reverse reaction and overall equilibrium reaction; and uses correct notation. The student has also successfully illustrated the reaction occurring in a sealed container, which is necessary for an equilibrium reaction to occur.

However, the ratio of reactant molecules to product molecules at equilibrium is incorrect. The ratio should be 1 : 2, not 1 : 1, according to the chemical equation. The ratio of 1 : 1 suggests that the concentrations of the reactants and products are equal at equilibrium, which is incorrect. At equilibrium, the rates of the forward and reverse reactions are equal; this is not shown. Additionally, the mix of reactants and products at equilibrium is not shown.

Overall, the student's model does not sufficiently represent dynamic equilibrium.

Mod 5 Equilibrium and Acid Reactions (CH12–5, 12–12)	Bands 5–6

Criteria	Marks
• Identifies at least THREE strengths of the model. AND • Identifies at least THREE limitations of the model. AND • Makes a judgement of the model.	6–7
• Identifies at least TWO strengths of the model. AND • Identifies at least TWO limitations of the model. AND • Makes a judgement of the model.	4–5
• Identifies at least ONE strength of the model. AND • Identifies at least ONE limitation of the model. AND • Makes a judgement of the model.	2–3
• Provides some relevant information	1

(b) *For example:*

The product yield could be increased by decreasing the pressure. According to Le Châtelier's principle, a decrease in pressure will shift the position of equilibrium to favour the reaction that produces more moles of gas. The reactant and product are gases with a ratio of 1 : 2; therefore, decreasing the pressure will increase the yield of the product.

Mod 5 Equilibrium and Acid Reactions (CH12–6, 12–12)	Bands 3–4

Criteria	Marks
• Identifies an appropriate method. AND • Provides justification using Le Châtelier's principle	2
• Identifies an appropriate method	1

Question 24

(a) *For example:*

Slice red cabbage into small pieces. Place the cabbage pieces and a small volume of water into a mortar and crush the cabbage pieces with a pestle. Once the mixture appears deep purple, use a pipette to extract the liquid (the indicator) from the mixture.

Add a few drops of a solution under investigation to the extracted indicator and note any colour change. The indicator will change to red or pink for acidic solutions, remain purple for neutral solutions or change to green or yellow for basic solutions.

Mod 6 Acid/Base Reactions (CH12–6, 12–13)	Bands 4–5

Criteria	Marks
• Uses an appropriate example. AND • Outlines the preparation. AND • Outlines testing the indicator using known acidic, basic and neutral solutions. AND • Refers to appropriate equipment	4
• Uses an appropriate example. AND • Outlines the preparation. AND • Outlines testing the indicator using known acidic, basic and neutral solutions	3
• Uses an appropriate example. AND • Outlines the preparation. OR • Outlines testing the indicator using known acidic, basic and neutral solutions	2
• Provides some relevant information	1

(b) Indicators are weak acids, which are represented by the symbol HIn. When an indicator is dissolved in water, it dissociates and reaches equilibrium with its conjugate base, In^-, according to the equation shown. The two species make the solution appear as different colours.

$$\underset{\text{colour I}}{HIn(aq)} + H_2O(l) \rightleftharpoons \underset{\text{colour II}}{In^-(aq)} + H_3O^+(aq)$$

The addition of an acid or a base to this system changes the position of equilibrium and, therefore, causes the indicator to change colour. Adding an acid increases the concentration of H_3O^+ in the system, causing the position of equilibrium to shift to the left – according to Le Châtelier's principle – and colour I to predominate. On the other hand, adding a base increases the concentration of OH^- in the system, causing the position of equilibrium to shift to the right and colour II to predominate.

Mod 6 Acid/Base Reactions (CH12–12, 12–13)	Bands 5–6
• Explains how an indicator dissociates in water. AND • States that the acid and conjugate base produce different colours. AND • Explains what happens when an acid is added to the indicator system AND what happens when a base is added to the indicator system. AND • Includes a relevant balanced chemical equation	4
• Any THREE of the above points	3
• Any TWO of the above points	2
• Provides some relevant information	1

 HSC_Chem_Y12_SS_2025

Question 25

(a) A buffer solution keeps pH relatively constant when small quantities of an acid or base are added to a system.

Mod 6 Acid/Base Reactions (CH12–6, 12–13)	Bands 3–4

- Outlines the function 1

(b) The research assistant should use methanoic acid *(weak acid)* and sodium methanoate *(conjugate base)*.

Methanoic acid dissociates slightly according to the equation:

$HCOOH(aq) \rightleftharpoons HCOO^-(aq) + H^+(aq)$ (equation 1)

Therefore, HCOOH predominates.

Sodium methanoate dissociates completely according to the equation:

$HCOONa(aq) \rightleftharpoons HCOO^-(aq) + Na^+(aq)$ (equation 2)

Therefore, $HCOO^-$ predominates.

When a small amount of an acid is added to the acidic buffer solution, the concentration of H^+ will increase. H^+ will react with $HCOO^-$ and shift the equilibrium position of equation 1 to the left, producing HCOOH. This will result in no change in pH.

When a small amount of a base is added to the acidic buffer solution, the concentration of OH^- will increase. OH^- will react with H^+ to form H_2O, causing more H^+ to dissociate from HCOOH and the equilibrium position of equation 1 to shift to the right. This will result in no change in pH.

Mod 6 Acid/Base Reactions (CH12–12, 12–13)	Bands 5–6

- Identifies BOTH solutions.
  AND
- Explains how the acidic buffer solution will work.
  AND
- Includes relevant balanced chemical equations........ 6–7

- Identifies BOTH solutions.
  AND
- Describes how an acidic buffer solution works.
  AND
- Includes a relevant chemical equation 4–5

- Identifies BOTH solutions.
  AND
- Outlines how an acidic buffer solution works........ 2–3

- Provides some relevant information 1

Question 26

For example:

Strong and weak refer to the relative dissociation of acids. Strong acids dissociate completely when dissolved in water, resulting in non-equilibrium systems. Hydrochloric acid is a strong acid and fully dissociates according to the equation $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$. Weak acids do not dissociate completely when dissolved in water, resulting in equilibrium systems. Ethanoic acid is a weak acid and does not fully dissociate according to the equation $CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$.

Concentrated and dilute refer to the amount of solute dissolved in solution. Concentrated acids contain a large amount of solute dissolved in solution; for example, 10 mol L^{-1}. Dilute acids contain a small amount of solute dissolved in solution; for example, 0.1 mol L^{-1}.

Note: Responses must explain the term dilute in terms of acid concentration, not in terms of adding water to a solution to decrease concentration.

Mod 6 Acid/Base Reactions (CH12–13)	Bands 5–6
• Explains all FOUR terms using relevant examples	4
• Explains THREE terms using relevant examples	3
• Explains TWO terms. OR • Explains ONE term using a relevant example	2
• Explains ONE term. OR • Provides some relevant information	1

Question 27

(a) $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$

Mod 7 Organic Chemistry (CH12–14)	Bands 2–3
• Provides the correct balanced chemical equation	1

(b) *For example:*

Prepare a fermentation mixture by placing 10.0 g of sugar, 5.00 g of dry yeast and 100 mL of water into a 250 mL conical flask. Loosely pack cotton wool into the top of the conical flask. Place the conical flask onto a top loading balance and record its initial mass. Reweigh the conical flask at regular intervals and record the time and mass. Keep the temperature constant. Plot the results in a graph.

Mod 7 Organic Chemistry (CH12–2, 12–4, 12–14)	Bands 4–5
• Outlines a relevant experiment	3
• Outlines a relevant experiment with ONE omission or error	2
• Provides some relevant information	1

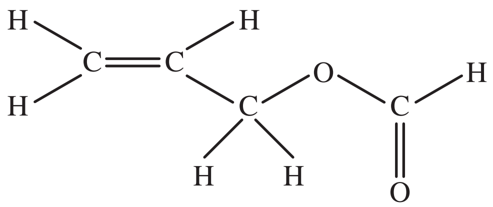
Question 28

(a) *For example, any two of:*

- using a heating mantle or a hot water bath as the heating source *(Compounds A and C are flammable. A heating mantle or hot water bath reduces the risk of fire.)*
- wearing safety goggles and gloves *(Compound B and H_2SO_4 are acids, which are corrosive. Goggles and gloves minimise the risk of eyes and skin coming into contact with these substances.)*
- heating the reaction under reflux conditions in a fume hood *(Organic vapours are irritants. Reflux conditions and a fume hood prevent organic vapours from entering the laboratory.)*

Mod 7 Organic Chemistry (CH12–2, 12–3, 12–14)	Bands 3–4
• Identifies TWO safe work practices	2
• Identifies ONE safe work practice.	1

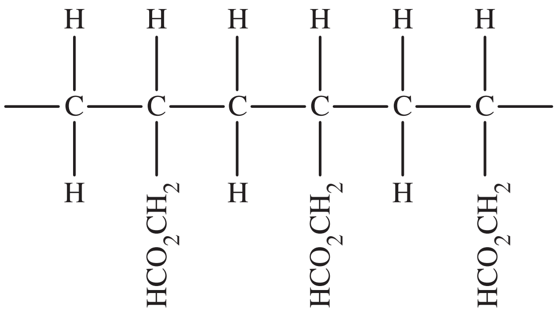
(b)



Mod 7 Organic Chemistry (CH12–14)	Bands 3
• Draws the structural formula of compound *C*	1

(c) An addition polymer can be produced from compound *C*. It can be produced by *(any one of)*:

- heating the monomer under pressure
- heating the monomer in the presence of a catalyst *(either a metallocene or Ziegler–Natta catalyst).*



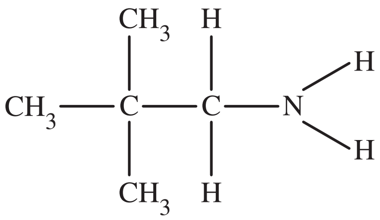
NOTE: Responses are only required to outline one method of production. Trimers with an addition polymer based on the answer to ***Question 28(b)*** *are acceptable.*

Mod 7 Organic Chemistry (CH12–6, 12–7, 12–14)	Bands 4–5
• Identifies that the type of polymer that can be produced is an addition polymer. AND • Outlines ONE possible method of production. AND • Draws the structural formula of three monomer units	3
• Any TWO of the above points	2
• Any ONE of the above points	1

Question 29

Name: 2,2-dimethylpropane-1-amine

Structure:



Infrared spectrum:

- The pair of peaks between 3300–3500 cm^{-1} are consistent with an amine (NH_2) group.
- The absence of a peak between 2220–2600 cm^{-1} rules out the presence of a nitrile group (C≡N).

^{1}H NMR spectrum:

- The 9H singlet at 0.95 ppm is indicative of the partial structure $(CH_3)_3C$–.
- The 2H singlet at 2.40 ppm is indicative of the partial structure –CH_2–N.
- The broad 2H singlet at 1.30 ppm is indicative of –NH_2.

^{13}C NMR spectrum:

- The three peaks indicate three unique carbon environments.

Mass spectrum:

- The molecular ion has a mass-to-charge (m/z) value of 87.
- The peak at m/z = 72 represents the loss of 15 amu from the molecular ion, indicating the loss of a methyl fragment (CH_3^+).
- The parent ion at m/z = 30 is consistent with a $CH_2NH_2^+$ fragment.

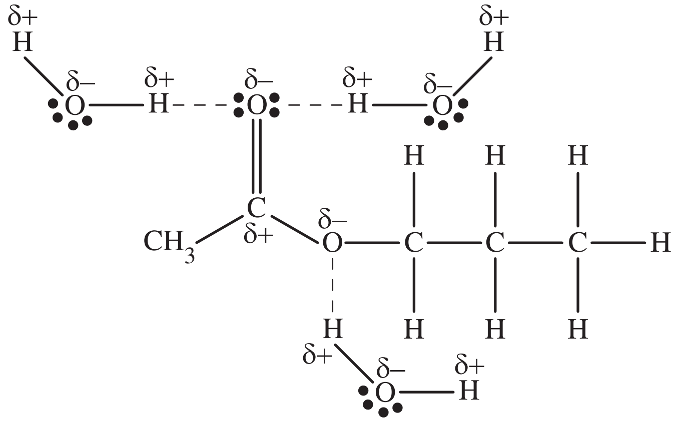
The spectrographic evidence suggests partial structures of $(CH_3)_3C$– and –CH_2NH_2. Therefore, the compound is 2,2-dimethylpropane-1-amine.

Mod 8 Applying Chemical Ideas (CH12–1, 12–2, 12–4, 12–6, 12–14, 12–15)	Bands 5–6

Criteria	Marks
• Identifies the compound. AND • Draws the structural formula. AND • Provides justification with reference to specific and relevant evidence from all FOUR spectra	6–7
• Identifies the compound. AND • Draws the structural formula. AND • Provides justification with reference to some evidence from at least TWO of the spectra.	4–5
• Identifies a compound OR draws the structural formula of a compound. AND • Attempts to provide justification with reference to evidence from any of the spectra	2–3
• Provides some relevant information	1

Question 30

(a)



Note: Responses are not required to show charges or lone pairs of electrons to obtain full marks.

Mod 7 Organic Chemistry (CH12–7, 12–14)	Bands 5–6
• Draws THREE hydrogen bonds between the hydrogens of water molecules and oxygens of the 1-propyl ethanoate molecule	3
• Draws TWO hydrogen bonds between the hydrogens of water molecules and oxygens of the 1-propyl ethanoate molecule	2
• Draws ONE hydrogen bond between the hydrogen of a water molecule and an oxygen of the 1-propyl ethanoate molecule	1

(b) Ethanol has a non-polar tail, which forms weak dispersion forces with 1-propyl ethanoate. It also has a polar OH group, which can form dipole–dipole forces and hydrogen bonds with 1-propyl ethanoate, and hydrogen bonds with water. This means that ethanol can increase the solubility of 1-propyl ethanoate.

Mod 7 Organic Chemistry (CH12–2, 12–3, 12–14)	Bands 3–4
• Identifies that ethanol has a non-polar tail and a polar OH group. AND • Explains that ethanol forms intermolecular bonds with 1-propyl ethanoate and water. AND • Explains that ethanol increases the solubility of 1-propyl ethanoate	3
• Describes at least TWO intermolecular bonds that ethanol forms with 1-propyl ethanoate and water	2
• Provides some relevant information	1

Question 31

Both processes use the same organic starting material and are conducted at similar low temperatures. However, process 1 uses oxygen (O_2) as the oxidising agent and an iron catalyst, which is less expensive than the oxidising agent used in process 2 (CrO_2Cl_2). Therefore, from an economic perspective, process 1 is preferable.

CrO_2Cl_2 is carcinogenic and, in process 2, produces chromium salts as by-products. Chromium is a heavy metal, and heavy metal residues can harm the environment. Water is the only by-product in process 1, meaning that from an environmental perspective, process 1 is preferable.

Process 2 converts a significantly higher percentage of toluene to benzaldehyde than process 1. Process 1 would require a recycling process to ensure that a higher percentage of toluene is converted. Therefore, from a yield perspective, process 2 is preferable.

Overall, process 1 is preferable.

Mod 8 Applying Chemical Ideas (CH12–4, 12–7, 12–14, 12–15)	Bands 5–6

- Identifies that process 1 is preferable.
  AND
- Provides justification with specific and relevant reference to the information provided 4

- Identifies that process 1 or process 2 is preferable.
  AND
- Provides some justification with reference to the information provided. 3

- Identifies that process 1 or process 2 is preferable.
  AND
- Attempts to provide justification with reference to the information provided 2

- Provides some relevant information 1

Question 32

(a) Converting the mass of butane combusted into moles gives:

$$\begin{aligned} n\left(\text{butane, } C_4H_{10}\right) &= \frac{m}{MM} \\ &= \frac{18.0}{4\times 12.01+10\times 1.008} \\ &= 0.30\ ...\ \text{mol} \end{aligned}$$

Calculating the heat released by the butane combusted gives:

$$\begin{aligned} \Delta H_c &= \frac{-q}{n} \\ -2878 &= \frac{-q}{0.30\ ...} \\ q &= 891.32\ \text{kJ} \\ &= 891\,328.28\ ...\ \text{J} \end{aligned}$$

Calculating the heat absorbed by the water gives:

$$\begin{aligned} q\times\frac{80}{100} &= mc\Delta T \\ 891\,328.28\ ...\times\frac{80}{100} &= 2150\times 4.18\times\Delta T \\ \Delta T &= 79.3\ ...°\text{C} \end{aligned}$$

Calculating the final temperature of the water gives:

$$\begin{aligned} T_{\text{final}} &= 79.3\ ...+12.0 \\ &= 91.3°\text{C} \end{aligned}$$

Mod 7 Organic Chemistry (CH12–4, 12–6, 12–14)	Bands 3–4

- Converts the amount of butane combusted into mol.
  AND
- Calculates the heat released by the butane.
  AND
- Calculates the heat absorbed by the water.
  AND
- Calculates the final temperature of the water 4

- Converts the amount of butane combusted into mol.
  AND
- Calculates the heat released by the butane.
  AND
- Calculates the heat absorbed by the water........ 3

- Converts the amount of butane combusted into mol.
  AND
- Calculates the heat released by the butane 2

- Provides some relevant working 1

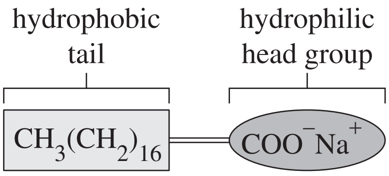
(b) *For example, any one of:*

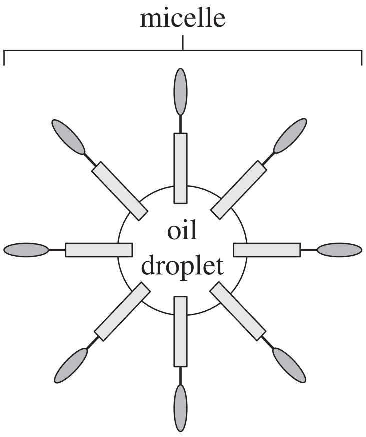
- Mining fossil fuels damages land, which increases soil erosion and results in loss of habitats.
- Mining fossil fuels releases methane (a greenhouse gas) into the atmosphere, causing pollution.
- Mining fossil fuels can result in environmental accidents, such as explosions, which can cause damage to land.
- Complete combustion of fossil fuels releases carbon dioxide (a greenhouse gas) into the atmosphere, leading to an increase in global temperatures and acidification of oceans.
- Incomplete combustion of fossil fuels produces carbon monoxide and soot, which are toxic to animals and humans.
- Burning fossil fuels that contain sulfur produces sulfur dioxide, which is a potent environmental pollutant.

Mod 7 Organic Chemistry (CH12–7, 12–14)	Bands 4–5
• Describes ONE environmental impact 1	

Question 33

Soap causes oil to form an emulsion with water. A soap molecule consists of a non-polar tail (lengthy hydrocarbon chain) and a polar head group (sodium salt of a carboxylic acid). The non-polar tail is hydrophobic and forms dispersion forces with oil molecules. This results in a micelle forming, in which the oil is trapped, and the hydrophilic polar head group is on the surface of the micelle. The polar head group forms dipole–dipole forces with water molecules, which enable the micelle to be suspended in water.





Mod 7 Organic Chemistry (CH12–7, 12–13, 12–14)	Bands 5–6

Criteria	Marks
• Explains the structure and polarity of a soap molecule. AND • Explains how a soap molecule forms bonds with oil and water. AND • Draws a relevant diagram	3
• Outlines how a soap molecule forms bonds with oil and water. AND • Draws a relevant diagram	2
• Provides some relevant information	1

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Question 34

For example:

Test 1: Add sodium chloride solution to the 1.00 mol L^{-1} solution. If Ag^+ is present, a precipitate of silver chloride will form, according to the equation $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$. If a precipitate forms, filter the mixture to obtain the filtrate.

Test 2: Add sodium hydroxide solution to the filtrate obtained in test 1. If Ca^{2+} is present, a precipitate of calcium hydroxide will form, according to the equation $Ca^{2+}(aq) + 2OH^-(aq) \rightarrow Ca(OH)_2(s)$. If a precipitate forms, filter the mixture to obtain the filtrate.

Test 3: Add sodium sulfate solution to the filtrate obtained in test 2. If Ba^{2+} is present, a precipitate of barium sulfate will form, according to the equation $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$.

Mod 8 Applying Chemical Ideas (CH12–7, 12–15)	Bands 5–6
• Outlines THREE tests. AND • Describes the expected observations for each test. AND • Provides a balanced chemical equation for each test	5
• Outlines THREE tests. AND • Describes the expected observations for each test. OR • Provides a balanced chemical equation for each test	4
• Outlines THREE tests	3
• Outlines TWO tests	2
• Provides some relevant information	1